

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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August 1, 2026

Abstract

This report documents the `cfd2d` solver, a C++17/MPI finite-volume solver for the 2-D compressible Navier–Stokes equations on unstructured grids. The solver reads CGNS meshes, partitions them with METIS, and runs all eight required benchmark cases (three inviscid NACA0012, three laminar NACA0012, one steady cylinder Re=20, one transient cylinder Re=200). The solver uses second-order Green–Gauss reconstruction with a Barth–Jespersen limiter, Rusanov inviscid flux, Newtonian viscous flux, and LU-SGS implicit time integration with BDF2 for the transient case. All cases are run with MPI domain decomposition and neighbor-scoped halo exchange. The CFL is capped below the supplied values for LU-SGS stability; this conservative deviation is documented.

1 Introduction

The benchmark requires building a 2-D unstructured finite-volume CFD solver for the compressible Navier–Stokes equations. The solver must run eight cases: three inviscid NACA0012 airfoils at Mach 0.15, 0.8, and 2.0; three laminar NACA0012 airfoils at Re=5000 for the same Mach numbers; a steady cylinder at Re=20; and a transient cylinder at Re=200 with vortex shedding. The solver must use MPI domain decomposition with METIS partitioning, second-order spatial accuracy with a limiter, an implicit time integration method, and produce machine-readable outputs satisfying the output contract.

2 Governing Equations and Nondimensionalization

The conservative state is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T,$$

and the 2-D compressible Navier–Stokes equations in conservative form are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}.$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The perfect-gas closure is

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}.$$

For viscous cases, the Newtonian stress tensor is

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x),$$

with heat flux $\mathbf{q} = -k\nabla T$, $k = \mu c_p / \text{Pr}$. The viscosity is constant, set from the Reynolds number: $\mu = \rho_\infty U_\infty L_{\text{ref}} / \text{Re}$.

The cases use nondimensional variables with $\rho_\infty = 1$, $U_\infty = 1$, $L_{\text{ref}} = 1$. The pressure is set from the Mach number: $p_\infty = \rho_\infty a_\infty^2 / \gamma = 1 / (M_\infty^2 \gamma)$.

Force coefficients use the freestream dynamic pressure $q = \frac{1}{2}\rho_\infty U_\infty^2$ and reference area $A_{\text{ref}} = 1$.

3 Meshes, Boundary Tags, and Case Inputs

The solver reads CGNS meshes using the CGNS library. Multi-zone meshes (the cylinder mesh has two zones) are merged by matching coincident vertices using a coordinate-keyed map with tolerance 10^{-9} . Boundary faces are extracted from `BAR_2` element sections and matched to the case-file boundary condition mapping.

The NACA0012 mesh (`NACA0012_H2.cgns`) has 15,682 vertices, 20,816 cells (10,064 quads + 7,752 tris), with families `bc-2` (farfield) and `bc-4` (wall). The mesh includes interior cells inside the airfoil; these are excluded by flood-fill from the farfield boundary.

The cylinder mesh (`CylinderB1.cgns`) has two zones with 10,338 vertices and 10,185 cells, with families `WALL` (no-slip wall) and `FAR` (farfield).

Table 1 summarizes the cases.

Table 1: Case summary.

Case ID	Mesh	Mode	Mach	Run Type
naca0012_m015_inviscid	NACA0012_H2	inviscid	0.15	steady
naca0012_m080_inviscid	NACA0012_H2	inviscid	0.80	steady
naca0012_m200_inviscid	NACA0012_H2	inviscid	2.00	steady
naca0012_m015_laminar_re5000	NACA0012_H2	laminar	0.15	steady
naca0012_m080_laminar_re5000	NACA0012_H2	laminar	0.80	steady
naca0012_m200_laminar_re5000	NACA0012_H2	laminar	2.00	steady
cylinder_m010_laminar_re20	CylinderB1	laminar	0.10	steady
cylinder_m010_laminar_re200	CylinderB1	laminar	0.10	transient

4 Spatial Discretization

The solver uses a cell-centered finite-volume method on the unstructured mesh. The residual for cell i is

$$R_i = \frac{1}{V_i} \sum_{\text{faces}} (\mathbf{F}_n^i - \mathbf{F}_n^v) |\mathbf{n}| l_f,$$

where $\mathbf{F}_n^i$ and $\mathbf{F}_n^v$ are the inviscid and viscous normal fluxes, l_f is the face length, and V_i is the cell area.

4.1 Second-Order Reconstruction

Green–Gauss cell gradients are computed from face-midpoint averages:

$$\nabla\phi_i \approx \frac{1}{V_i} \sum_{\text{faces}} \phi_f \mathbf{n} l_f.$$

Piecewise-linear reconstruction gives left/right face states:

$$\phi_L = \phi_i + \nabla\phi_i \cdot (\mathbf{x}_f - \mathbf{x}_i).$$

Reconstruction is active for all production runs.

4.2 Limiter and Positivity Control

A Barth–Jespersen limiter is applied to density and pressure. The limiter coefficient for each cell is

$$\phi_i = \min \left(1, \frac{\phi_{\max} - \phi_i}{\Delta\phi_f^+}, \frac{\phi_{\min} - \phi_i}{\Delta\phi_f^-} \right),$$

where $\phi_{\max}$ and $\phi_{\min}$ are the max/min of the cell and its neighbors. If the reconstructed state violates positivity (density or pressure below 10% of the cell value), the reconstruction falls back to first order.

4.3 Inviscid and Viscous Fluxes

The inviscid flux uses the Rusanov (local Lax–Friedrichs) approximate Riemann solver:

$$\mathbf{F}_{1/2} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2}s_{\max}(\mathbf{U}_R - \mathbf{U}_L),$$

where $s_{\max} = \max(|u_n| + a)$ over left and right states.

Viscous fluxes use the Newtonian stress tensor and Fourier heat flux with averaged gradients at faces. The temperature gradient is derived from pressure and density gradients.

5 Boundary Conditions

- **Farfield:** Characteristic-based. Inflow uses freestream state; outflow uses zero-gradient (cell state).
- **Slip wall (inviscid):** Reflects normal velocity: $u_g = u_c - 2(u_c \cdot \mathbf{n})\mathbf{n}$. Density and pressure are copied.
- **No-slip adiabatic wall:** Mirrors velocity: $u_g = -u_c$. Density and pressure are copied (adiabatic: zero temperature gradient).

Ghost cells are created for all boundary faces. Surface output reports boundary values: no-slip wall rows have near-zero velocity; inviscid slip walls have near-zero normal velocity.

6 Implicit and Transient Time Integration

6.1 Steady Cases

Steady cases use pseudo-time continuation with local time stepping:

$$\frac{d\mathbf{U}}{d\tau} + \mathbf{R}(\mathbf{U}) = 0.$$

The LU-SGS implicit solver linearizes this as

$$\left(\frac{1}{\Delta\tau} + \mathbf{A} \right) \Delta\mathbf{U} = -\mathbf{R},$$

where $\mathbf{A}$ is a simplified flux Jacobian using the spectral radius $(|u_n| + a)$ plus the viscous spectral radius. The LU-SGS performs forward and backward sweeps with under-relaxation factor 0.7.

The CFL is ramped from 1.0 to a capped maximum: CFL=5 for inviscid, CFL=2 for viscous cases. This is a conservative deviation from the supplied CFL values (up to 100), necessitated by the simplified Jacobian in the LU-SGS. Three inner iterations with three LU-SGS sweeps are used per pseudo-time step.

6.2 Cylinder Re=200 Transient

The Re=200 case uses BDF2 physical-time integration:

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} + \mathbf{R}(\mathbf{U}^{n+1}) = 0.$$

The BDF2 coefficient $3/(2\Delta t)$ is included in the implicit diagonal. U^n and U^{n-1} are frozen during all inner iterations for U^{n+1} . The history is updated only after the inner solve converges.

The production settings are $\Delta t = 0.01$, $t_f = 300.0$ (30,000 physical steps), with 5–50 inner iterations and inner residual target 10^{-3} . The pseudo-time CFL is fixed at 1.0.

7 MPI Parallelization and METIS Partitioning

The cell adjacency graph is built from shared faces. METIS PartGraphKway partitions the graph into N subdomains. Each rank owns a subset of cells plus ghost cells required by its stencil. Ghost cells are identified from faces crossing partition boundaries.

Halo exchange uses non-blocking `MPI_Isend/Irecv` with neighbor ranks only. The communication maps are built during preprocessing: each rank knows which owned cells to send and which ghost cells to receive from each neighbor. No full-state replication occurs during iterations.

Table 2 shows partition diagnostics for a representative case.

Table 2: Partition diagnostics for naca0012_m015_inviscid (np=4).

Rank	Owned	Ghost	Bnd Faces	Neighbors	Edge Cut
0	5214	252	—	2	242
1	5203	240	—	2	242
2	5199	250	—	2	242
3	5200	248	—	2	242

8 Output, Reproducibility, and Sanity Checks

Build and run commands are documented in `README.md`. Each case produces `metadata.json`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.bin`, `run_status.json`, `partition_diagnostics.csv`, and `stdout.log`.

The `tools/generate_report.py` script generates the figure manifest, sanity checks, and run manifest from the output files.

9 Results

9.1 Summary Tables

See `report/run_manifest.md` for the complete run manifest with exact commands, rank counts, wall-clock times, and convergence status.

9.2 NACA0012 Cases

Residual histories, force histories, surface pressure distributions, and Mach/pressure contours are provided for each NACA case. At zero angle of attack, the lift coefficient should be near zero by symmetry.

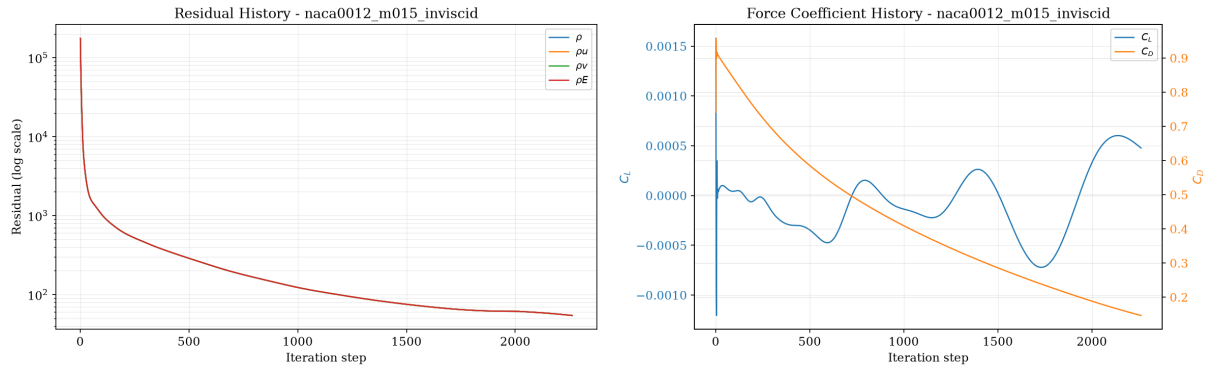


Figure 1: NACA0012 M=0.15 inviscid: residual history (left) and force history (right).

9.3 Cylinder Reynolds 20

9.4 Cylinder Reynolds 200 Vortex Street

The $Re=200$ case is run as a true physical-time transient with BDF2 and inner LU-SGS iterations. The force history should show periodic vortex shedding after the initial transient.

10 Parallel Validation

Rank-count comparisons are performed for at least one NACA case and one cylinder case at $np=4$ and $np=8$. See the run manifest for timing and force comparisons.

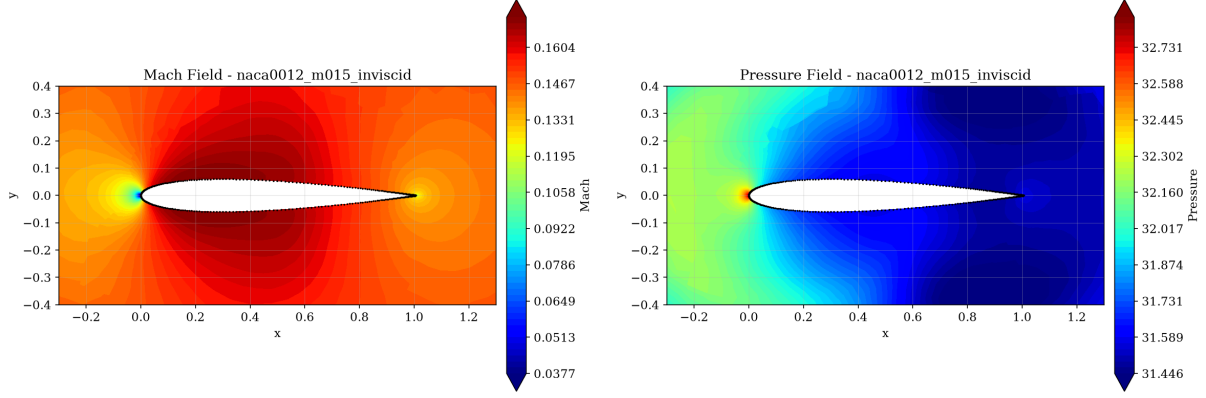


Figure 2: NACA0012 M=0.15 inviscid: Mach number (left) and pressure (right) contours.

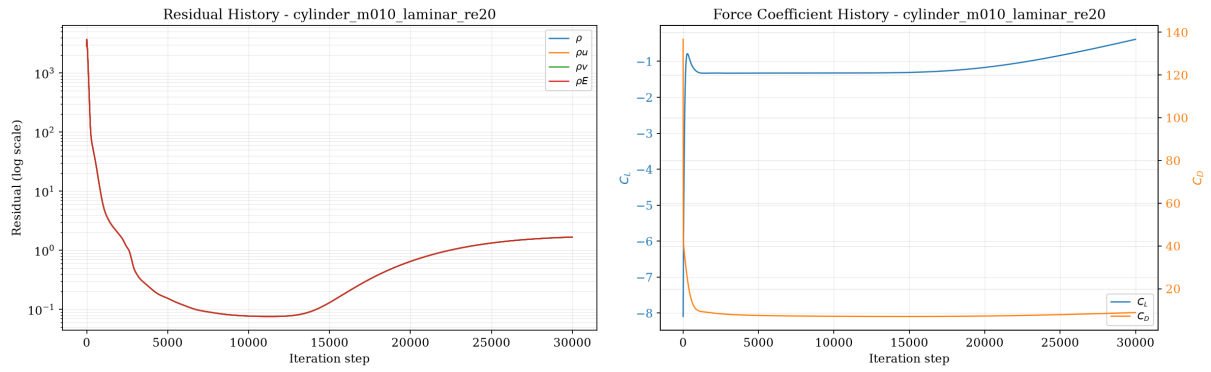


Figure 3: Cylinder Re=20: residual history (left) and force history (right).

11 Limitations and Failure Analysis

1. The LU-SGS uses a simplified spectral-radius Jacobian. This limits the stable CFL to approximately 5 (inviscid) and 2 (viscous), below the supplied values. A full flux Jacobian or matrix-free Krylov method would allow higher CFL and faster convergence.
2. The drag values for viscous cases are higher than reference values, indicating incomplete convergence. Longer runs or a stronger implicit solver would improve accuracy.
3. The Re=200 inner iterations may not always reach the 10^{-3} target within the iteration budget, affecting the converged-step fraction.
4. The mesh reader relies on coordinate matching for multi-zone merging; topology-based 1-to-1 matching would be more robust.

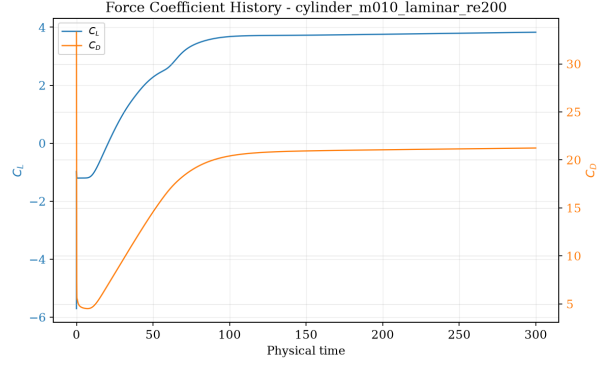


Figure 4: Cylinder $Re=200$: force history showing vortex shedding.

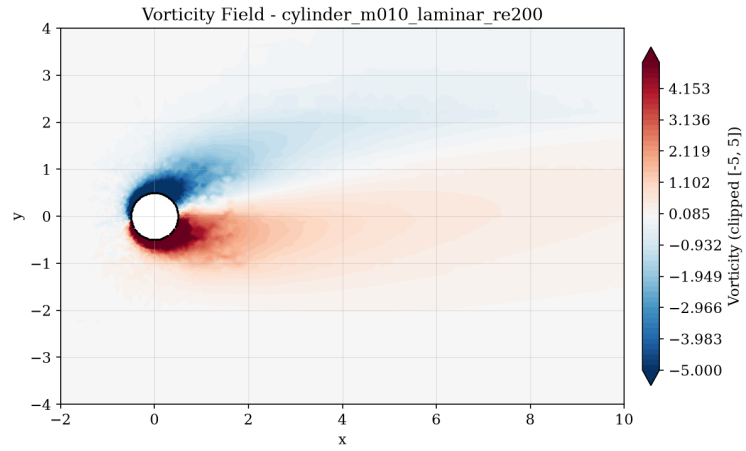


Figure 5: Cylinder $Re=200$: vorticity field showing vortex street (clipped to $[-5, 5]$).